

FogGuard AI

Proactive Safety System for Mine Vehicles in Fog and Low Visibility

Problem statement: SIH26007 — Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.

Executive concept

FogGuard AI is a low-cost, vehicle-mounted safety and decision-support system for manually driven mine vehicles. Instead of relying only on obstacle detection after a vehicle enters fog, it measures local atmospheric risk, estimates usable visibility, detects hazards using fog-resilient sensing, and shares a live hazard/fog-risk map with nearby vehicles and the mine supervisor.

The core idea is to treat fog as a **dynamic operational hazard layer** on the mine map. The system recommends a safe speed and route before a vehicle enters a high-risk zone, while retaining local collision warning and emergency-stop outputs if an obstacle becomes imminent.

What makes it distinct

Most conventional systems focus on one or more of these approaches: camera/LiDAR perception, radar-based collision warning, proximity alarms, autonomous haulage, or vehicle positioning. FogGuard AI combines these into a practical decision layer that is specifically designed for low-cost retrofit deployment:

- Measures environmental degradation locally rather than inferring fog solely from a camera image.
- Converts atmospheric measurements into a vehicle-specific visibility/risk estimate.
- Uses radar as the primary long-range safety sensor in fog; low-cost optical/thermal/depth sensing becomes supplementary rather than the only safety dependency.
- Builds a shared map of fog-risk and detected hazards from multiple vehicles.
- Provides a fail-safe speed policy: as sensing confidence falls, permitted speed falls; if confidence becomes insufficient, the system commands a controlled stop or requests manual confirmation.
- Is designed as a retrofit add-on for existing dumpers, water tankers, light vehicles, excavators, and service vehicles.

System objectives

1. Detect vehicles, workers, berms, rocks, and other obstacles in reduced visibility.
2. Estimate visibility and sensor reliability in real time.
3. Calculate a safe operating speed using stopping distance, road condition, grade, and sensed visibility.
4. Warn a driver and supervisor before a vehicle enters high-fog or high-risk segments.
5. Exchange risk messages between vehicles without requiring continuous internet connectivity.
6. Record events for post-incident review and safety compliance.
7. Maintain safe operation during partial sensor, network, or GNSS failure.

Three-layer architecture

1. Hardware layer

Vehicle unit

Module	Recommended prototype component	Role	Why it is useful
Edge controller	Raspberry Pi 5 / Orange Pi 5	Runs fusion, UI, logging, communication	Keeps critical perception local; does not depend on cloud latency
Long-range sensing	60/77 GHz mmWave radar development board	Range, relative speed, bearing of obstacles	Radio sensing remains more usable than visible cameras in fog, dust, darkness, and glare
Short-range depth	VL53L5CX multi-zone ToF sensor or compact depth module	Near-field spatial confirmation	Covers close obstacles and radar minimum-range limitations
Thermal sensor	MLX90640 or compact thermal camera	Human/vehicle heat-pattern cue	Provides a secondary cue in darkness and visually degraded conditions
Air-risk sensing	Temperature-humidity-pressure sensor plus particulate sensor	Estimate fog/dust likelihood and atmospheric condition	Enables the system to measure environmental degradation, not only detect obstacles
Close-range safety	Four ultrasonic sensors	0.2–4 m obstacle detection	Helps during reversing, loading, and slow maneuvering
Positioning	GNSS receiver; optional IMU	Mine-map position, heading, dead reckoning	Enables route risk alerts and records where hazards occur
Vehicle interface	CAN/OBD interface or isolated relay prototype	Read speed; issue advisory/controlled-stop output	Supports real vehicle integration while keeping a demo safe
Driver alerts	7-inch display, buzzer, vibration motor, RGB LED strip	Clear multi-level alerting	Alert remains understandable even when the driver cannot look at a screen

Module	Recommended prototype component	Role	Why it is useful
Vehicle communications	LoRa / Wi-Fi mesh / ESP-NOW prototype link	Exchange risk messages vehicle-to-vehicle	Allows local operation when cellular coverage is unreliable
Data storage	SD card / local SQLite database	Event logging	Supports audit, model improvement, and incident reconstruction

Prototype hardware layout

Roof / front mount

- └─ mmWave radar: forward field of view
- └─ Thermal camera: forward field of view
- └─ ToF/depth sensor: forward near field
- └─ GNSS antenna and IMU
- └─ Environmental sensor inlet protected by a breathable filter

Bumper / rear / side mounts

- └─ Ultrasonic sensors for close-range warning

Cabin enclosure

- └─ Raspberry Pi / edge processor
- └─ 7-inch driver display
- └─ buzzer, vibration warning and LEDs
- └─ local SD logging
- └─ LoRa or Wi-Fi mesh communication module

Real-life ruggedization

- Use an IP65/IP67 enclosure, cable glands, automotive connectors, fused DC-DC buck converter, reverse-polarity protection, and TVS protection.
- Mount sensors on vibration-isolated brackets and use a protective radar radome.
- Place the environmental sensor away from exhaust and direct wheel spray; use replaceable inlet filters.
- Use a watchdog timer and health-check LEDs so an operator can see whether the system is ready.
- Prototype automatic braking as an isolated relay/LED demonstration only; actual vehicle braking integration must require OEM approval, functional-safety engineering, and mine authorization.

2. AI and decision layer

A. Visibility and atmosphere risk estimator

Inputs: relative humidity, temperature, particulate level, rain/wiper state if available, camera contrast score, radar return quality, and time/location.

Outputs: estimated visibility band, environmental risk score, sensor confidence values, and operating mode.

Example operating modes:

Mode	Visibility / confidence condition	System behaviour
Green	Clear or good confidence	Normal monitoring and logging
Yellow	Moderate fog/dust or reduced confidence	Lower recommended speed; issue early caution
Orange	Low visibility	Strong warning; geofence high-risk segments; restrict speed
Red	Very low visibility or critical sensing uncertainty	Controlled stop recommendation; supervisor alert; route hold or safe pull-over guidance

A lightweight regression or classification model can be trained using labelled trials in clear air, artificial fog, dust, dawn/night conditions, and wet surfaces. For SIH, begin with a transparent rule-based estimator and then add a small ML model; judges can validate the reasoning more easily.

B. Reliability-aware sensor fusion

FogGuard AI does not assume every sensor is equally reliable at all times. It calculates a confidence score for each sensor and uses those values in fusion.

```
Obstacle confidence =  
  radar confidence × radar weight  
+ depth confidence × depth weight  
+ thermal confidence × thermal weight  
+ ultrasonic confidence × ultrasonic weight
```

In clear conditions, depth/camera evidence can have higher influence. In fog or dust, the radar weight rises while optical confidence is reduced. A Kalman filter tracks obstacle range and relative velocity over time, reducing one-frame false alarms.

Output per tracked object: object ID, range, bearing, estimated relative speed, class estimate, confidence, and collision risk.

C. Physics-based collision risk engine

The first version should use explainable physics rather than claim an unnecessarily complex deep model.

For each detected obstacle, compute time-to-collision:

$$TTC = \frac{d}{v_{relative}}$$

where d is closing distance and $v_{relative}$ is positive closing speed.

Compute conservative stopping distance:

$$d_{stop} = vt_r + \frac{v^2}{2a_{safe}} + d_{margin}$$

where v is vehicle speed, t_r is sensing/driver/system reaction time, a_{safe} is safe deceleration adjusted for road grade and surface condition, and d_{margin} is a fixed safety margin.

Trigger levels:

- **Caution:** available distance is approaching stopping distance.
- **Warning:** TTC is below a selected threshold or obstacle lies in projected path.
- **Critical:** braking distance is insufficient without immediate deceleration; activate buzzer/vibration and critical dashboard state.
- **Fail-safe:** if sensor confidence is critically low, impose a conservative speed limit or controlled stop recommendation.

D. Fog-risk map and cooperative intelligence

Every equipped vehicle periodically broadcasts a small message:

```
{
  "vehicle_id": "V12",
  "timestamp": "UTC time",
  "position": "mine-grid or GNSS coordinate",
  "speed_kmph": 18,
  "visibility_band": "orange",
  "risk_score": 0.78,
  "hazard": "obstacle / fog / stopped vehicle / road blockage",
  "confidence": 0.87
}
```

The local gateway merges reports into mine-grid cells, for example 20 m × 20 m. Each cell stores a decaying risk score so old reports gradually lose influence. A vehicle approaching an orange/red cell receives a route and speed warning before it physically enters the area.

E. Safe-speed recommender

The recommended speed is the maximum speed that maintains a conservative stopping distance within the reliable detection/visibility range.

```
Safe speed = function(
  usable visibility,
  radar detection confidence,
  road grade,
```

```
road-surface condition,  
loaded/unloaded state,  
vehicle braking model,  
curvature/intersection risk,  
obstacle risk  
)
```

For the prototype, use configurable vehicle profiles: light vehicle, loaded dumper, unloaded dumper. Never present a generic speed number as valid for all real haul trucks; the mine must calibrate braking performance and its operating rules.

3. Software layer

Driver dashboard

The driver interface must be glanceable and low-distraction.

- Large visibility state: Green, Yellow, Orange, or Red.
- Current speed and recommended maximum speed.
- Directional obstacle indicator: front-left, front, front-right, rear, or side.
- Range and urgency: for example, “Vehicle ahead — 18 m — close rate high.”
- Mine-route risk warning: “Fog-risk zone ahead in 120 m.”
- Simple action instruction: “Reduce to 15 km/h,” “Stop at safe bay,” or “Proceed with caution.”
- Sensor health indicator; avoid hiding a failed sensor.
- Audible cues with different patterns for caution, warning, and critical events.

Supervisor dashboard

A web dashboard for the mine safety office shows:

- Live vehicle positions on a simplified mine map.
- Fog-risk heat map and latest visibility reports.
- Active red/orange incidents and acknowledgement state.
- Vehicle sensor health and communication status.
- Replay of routes and alerts for incident analysis.
- Daily reports: exposure time in fog zones, overspeed warnings, near-miss events, and unavailable sensors.

Suggested stack: FastAPI backend, SQLite/PostgreSQL database, MQTT for messages, and a React/Streamlit frontend. For SIH, a laptop hotspot can host the gateway and dashboard locally.

Data flow

Sensors

- > Edge acquisition service
- > Sensor health and timestamp validation
- > Visibility/risk estimator
- > Reliability-aware fusion
- > Object tracking and collision-risk calculation
- > Driver alert + safe-speed recommendation
- > Local event logger
- > Mesh/MQTT risk broadcast
- > Supervisor fog map and fleet dashboard

Key features and their purpose

Feature	Practical purpose
Atmospheric-risk measurement	Identifies fog/dust deterioration early and supports visibility-aware operating mode
Radar-first perception	Maintains obstacle ranging and velocity estimation when vision degrades
Multi-sensor confidence fusion	Prevents blind trust in a sensor that is degraded, blocked, or faulty
Physics-based safe-speed calculation	Converts detection into an understandable and actionable instruction
Shared risk map	Warns approaching vehicles before they encounter the same hazard
Geofenced caution zones	Protects intersections, dumps, loading points, bends, and known fog pockets
Driver haptic/audio alerts	Maintains warning effectiveness when attention is on driving, not a display
Supervisor dashboard	Allows fleet-level response and safety auditing
Local offline operation	Keeps core collision warning active despite poor network coverage
Event recorder	Enables evidence, maintenance diagnostics, and iterative model tuning
Sensor health monitoring	Avoids a dangerous false assumption that the system is functioning normally

Prototype design for SIH

Demonstration vehicle

Use a 1:10 or 1:12 remote-control truck or a small robotic four-wheel platform. Mount the sensing pod on top and connect the driver display/dashboard to a laptop or tablet.

Demonstration mine model

Build a tabletop open-cast mine route with:

- A curved haul road and blind intersection.
- A parked vehicle or barrier.
- A small worker figurine with a warm thermal source behind/near it.

- A fog zone made with a sealed transparent acrylic chamber and a water-based ultrasonic humidifier.
- Two prototype vehicles or one vehicle plus a fixed “roadside node.”

Avoid smoke, incense, or combustion aerosols in an indoor demo because they can irritate people and distort the distinction between fog and dust.

Live demo sequence

1. Start in clear conditions: dashboard is green, normal speed is allowed.
2. Place an obstacle ahead: radar identifies range and closing speed; dashboard gives a directional caution/warning.
3. Introduce fog in the chamber: environmental and optical confidence change; dashboard moves to yellow/orange and recommended speed reduces.
4. Show adaptive operation: radar retains primary weight while optical/depth confidence is reduced.
5. Broadcast the fog-risk report from Vehicle A or a fixed node.
6. Show Vehicle B/supervisor dashboard receiving “high fog ahead” before reaching the chamber.
7. Demonstrate a critical obstacle: buzzer/vibration alert activates; vehicle stops or servo throttle is cut in the model.
8. Show logged incident data and a dashboard replay.

Bill of materials

The precise cost depends on the radar board selected. The table intentionally separates prototype and field-grade expectations.

Item	Prototype estimate (INR)	Notes
Raspberry Pi 5 / Orange Pi 5, power supply, SD card	6,000–9,000	Use an actively cooled case
mmWave radar development board	5,000–18,000	Choose a board with usable point/range/velocity output; this is the most important sensor
VL53L5CX ToF board	1,500–3,000	Near-field depth grid
MLX90640 thermal module	4,000–8,000	Better than an 8×8 thermal array for a convincing human demo
Temperature-humidity-pressure sensor	300–800	BME280/BME680 class
Particulate sensor	1,000–2,500	PMS5003/SPS30 class; interpret as air-risk input, not a direct fog meter
GNSS + IMU	1,000–3,000	Basic module adequate for tabletop/demo; RTK is a field upgrade
Ultrasonic sensors and mounts	600–1,500	Close-range redundancy

Item	Prototype estimate (INR)	Notes
LoRa/Wi-Fi mesh modules	1,000–3,000	Two nodes needed for cooperative demo
Display, buzzer, LEDs, vibration motor	2,000–5,000	Driver interaction
DC-DC converter, fuse, wiring, connectors	1,500–3,000	Do not omit electrical protection
Enclosure, brackets, 3D prints	2,000–5,000	Include vibration isolation
Expected prototype total	25,000–60,000	Can be reduced by using one vehicle and a fixed infrastructure node

Important correction on sensor selection

Do not use an RCWL-9600 as the main radar for this problem. It is only a basic microwave motion-presence module and does not provide dependable object range, angle, or relative velocity needed for collision avoidance. For a credible prototype, use a real FMCW mmWave radar development board capable of reporting range and Doppler/velocity.

Likewise, a low-resolution 8×8 thermal sensor can demonstrate heat presence but is weak for dependable person classification. Use it only as a secondary cue, or choose MLX90640/thermal camera if budget permits.

Implementation plan

Week 1: Architecture and sensor validation

- Finalize safety requirements, operating modes, and measurable success criteria.
- Integrate and test each sensor individually.
- Verify radar range, angle, update rate, and output format in fog-free conditions.
- Build a basic serial/MQTT data acquisition service and timestamp all readings.

Week 2: Edge safety functions

- Implement sensor health checks, obstacle tracking, TTC, stopping-distance logic, and multi-level alerts.
- Implement rule-based visibility risk classification.
- Build the driver dashboard and local event logger.

Week 3: Cooperative fog map

- Implement LoRa/Wi-Fi messages and a local MQTT gateway.
- Create the mine-grid risk map and supervisor dashboard.
- Add geofence segments: blind bend, intersection, loading zone, dump zone, and safe bay.

Week 4: AI improvement and validation

- Record labelled runs in clear, artificial fog, dust-like safe test conditions, daylight, and darkness.
- Train a lightweight model to estimate visibility/risk and sensor confidence.
- Compare rule-only versus confidence-aware fusion.
- Quantize/optimize the model if required for edge execution.

Week 5: Demonstration and documentation

- Build the tabletop mine and fog chamber.
- Run repeatable test scenarios and record evidence.
- Prepare a short demo video as backup.
- Document BOM, wiring, APIs, calibration procedure, limitations, and field deployment roadmap.

Evaluation metrics

Do not claim arbitrary accuracy numbers. Measure and show real prototype metrics.

Metric	How to measure
Obstacle detection rate	Correct detections divided by total test obstacles at specified distances and fog levels
False-alert rate	Number of false warnings per route/test minute
Range error	Difference between radar-estimated and tape-measured obstacle range
Alert latency	Time from obstacle appearance to driver alert
Communication delay	Time from node A fog report to node B/dashboard receipt
Safe-speed response	Time and distance required for recommendation to change after entering fog zone
Fog-map lead time	Seconds/metres of advance warning before second vehicle enters reported zone
Availability	Percentage of test time with valid sensor health and processing

Suggested test matrix:

- Clear daylight, clear low light, artificial fog, dust-safe simulation, static obstacle, moving obstacle, stopped vehicle around bend, sensor blocked/fault injected, and network disconnected.

Safety and real-world deployment

This prototype is a decision-support and collision-warning system, not an independently certified vehicle control system. A real deployment must be developed with mine management, vehicle OEMs, and qualified functional-safety engineers.

Mandatory field considerations:

- Validate radar performance around iron ore, metal structures, water spray, haul-road berms, and multiple heavy vehicles.
- Calibrate braking curves separately for each vehicle type, loading condition, gradient, tyre condition, and road surface.
- Define mine-specific rules for maximum speed, right-of-way, safe stopping bays, and red-zone entry.
- Use authenticated communication, device identity, encrypted messages where feasible, and role-based dashboard access.
- Provide manual override, diagnostic mode, fail-safe power design, and maintenance checks.
- Conduct staged pilots: test track -> one light vehicle -> one service vehicle -> limited fleet -> larger rollout.

Pitch

Existing systems mainly try to see through fog after a vehicle is already inside it. FogGuard AI turns fog into a live, shared operational risk map. It measures local atmospheric degradation, uses radar-first perception to keep detecting hazards, calculates an explainable safe speed, and warns vehicles before they enter the dangerous zone. It is a practical retrofit architecture for Indian open-cast mines: local safety continues offline, while cooperative mapping improves safety as more vehicles participate.

Final SIH strategy

To maximize competitiveness, do not position the project as “another AI collision detector.” Position it as a **proactive, cooperative low-visibility operating system** for mine traffic.

Your strongest proof points should be:

- A physical fog-chamber demonstration with a moving vehicle and actual sensor data.
- A measurable comparison: camera/depth confidence drops in fog while radar and the safety policy remain usable.
- Two-node cooperative warning: one node sees poor visibility; another gets advance warning.
- A transparent safe-speed calculation tied to stopping distance, not a vague AI recommendation.
- Honest limitations and a credible field-validation plan.
- A realistic BOM and ruggedization pathway, showing the project can move from SIH prototype to mine pilot.